



CLINICAL ARTICLE

Vaginal group B streptococcus status during intrapartum antibiotic prophylaxis[☆]Santiago Scasso^{a,*}, Joel Laufer^a, Grisel Rodriguez^b, Justo G. Alonso^a, Claudio G. Sosa^{a,c}^a Department of Obstetrics and Gynecology, Pereira Rossell Hospital, University of Uruguay, Montevideo, Uruguay^b Department of Microbiology, University of Uruguay, Montevideo, Uruguay^c Department of Epidemiology, University of Uruguay, Montevideo, Uruguay

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ABSTRACT

Objective: To assess maternal group B streptococcus (GBS) colonization status and the pharmacokinetic profile of penicillin G in the umbilical cord and amniotic fluid compartment during 4 hours of intrapartum antibiotic prophylaxis (IAP). **Methods:** In a prospective study at a hospital in Montevideo, Uruguay, 60 GBS carriers in active labor after a singleton pregnancy of 37 weeks or more were enrolled between April 1, 2011, and April 30, 2012. Intravenous penicillin G was administered via a standard regimen. Rectovaginal samples were obtained before IAP initiation, and 2 and 4 hours after the initial dose. Penicillin G concentrations were measured by high-performance liquid chromatography. Samples were obtained from fetal cord blood in all cases and from amniotic fluid obtained from patients who delivered by cesarean. **Results:** Among the 60 participants, 43 (72%) had a positive rectovaginal sample before IAP initiation. Of these women, 23 (53%) had negative cultures after 2 hours; after 4 hours, only 5 (12%) remained positive for GBS. The penicillin G concentration in amniotic fluid and cord blood was above the minimum inhibitory concentration (0.12 µg/mL) in all cases. **Conclusion:** Four hours of IAP was needed to reduce the number of women with positive GBS cultures to 12%. Therefore, 4 hours of IAP might be necessary to achieve overall effectiveness from this treatment.

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1. Introduction

Group B streptococcus (GBS) emerged as an important cause of perinatal morbidity and mortality in the 1970s and remains one of the leading causes of early-onset neonatal sepsis in the USA [1]. Between 10% and 30% of pregnant women carry GBS, which can be passed to the neonate during passage through the birth canal [2,3].

Vertical transmission of GBS during labor or delivery can result in early-onset GBS infection—an invasive infection that develops during the first week of life and has a mortality rate of almost 50% [4]. International guidelines and protocols set out in the 1990s have resulted in an 80% decrease in the incidence of early-onset GBS sepsis worldwide (from 1.7 to 0.4 cases per 1000 live births) [5–7]. This reduction is due to the implementation of a systematic, universal, culture-based strategy to identify GBS-carrying mothers between 35 and 37 weeks of pregnancy who are candidates for intrapartum antibiotic prophylaxis (IAP) [1,4].

The Centers for Disease Control and Prevention (CDC) recommends treating all GBS-carrying pregnant women with at least 4 hours of IAP to

sufficiently protect the newborn [4]. The main goals of IAP are to reduce the vaginal colony count at delivery, prevent GBS from ascending and proliferating in the amniotic fluid compartment, and attain sufficient antibiotic levels in the fetal bloodstream during labor with a minimum inhibitory concentration (MIC) of penicillin of 0.12 µg/mL [4,8,9]. Although IAP seems to reduce the incidence of early-onset GBS disease, Ohlsson and Shah [10] found that its effect needs to be further substantiated owing to a lack of high-quality evidence.

Guidelines recommend at least 4 hours of IAP as an adequate treatment, but the origin of this duration is not clear and a shorter time could be sufficient for neonatal management [4]. For example, De Cueto et al. [11] have suggested that 1 or 2 hours of IAP might effectively reduce neonatal GBS colonization or disease in women with established risk factors for early-onset GBS disease in the newborn. Another important issue is the fact that in clinical practice only 50% of GBS-carrying mothers generally receive the full 4 hours of IAP [12]. Furthermore, the most recent CDC guidelines state that no medically necessary obstetric procedure should be delayed to achieve 4 hours of GBS prophylaxis before delivery [4]. As a result, the key question is whether newborns who do not receive 4 hours of IAP have any protection against vertical transmission of GBS.

The primary aim of the present study was therefore to assess maternal rectovaginal colonization by GBS, specifically in the first 4 hours after penicillin administration, among GBS-carrying women in labor.

[☆] Preliminary results were presented at the 10th World Congress of Perinatal Medicine; November 8–11, 2011; Punta del Este, Uruguay.

* Corresponding author at: Orleans 2273 CP 11500 Montevideo, Uruguay. Tel.: +598 99277738; fax: +598 26000434.

E-mail address: sscasso@gmail.com (S. Scasso).

A secondary aim was to examine the pharmacokinetic profile of the maternal penicillin G (benzylpenicillin) dosing regimen in the umbilical cord blood and amniotic fluid during the first 4 hours after the initial maternal infusion.

2. Materials and methods

The present prospective study recruited GBS carriers in active labor admitted to a public obstetric service at Pereira Rossell Hospital, Montevideo, Uruguay, between April 1, 2011, and April 30, 2012. Healthy women were eligible for inclusion when they had a singleton pregnancy of 37 weeks or more, were GBS positive as diagnosed by rectovaginal culture between 35 and 37 weeks, and were in the active phase of labor. The exclusion criteria were precipitous deliveries in which it was not possible to collect samples after IAP, penicillin allergy, current use of antibiotics, and renal disease. The study protocol was approved by the Scientific Board of the Obstetric-Gynecology Department and the Ethics Committee of Uruguay University School of Medicine. All participants provided informed consent.

Intravenous penicillin G prophylaxis during labor was administered to every participant via the standard CDC dosing regimen (an initial infusion of 5 million units, followed by 2.5 million units every 4 hours until delivery) [4]. Data about the duration and timing of antibiotic infusion and biological samples were collected, together with maternal characteristics. A specific form and a Microsoft Access 2010 database (Microsoft, Redmond, WA, USA) were designed for data management.

A GBS rectovaginal sample, obtained in accordance with the 2010 CDC protocol using one swab [4], was collected from all women at the time of admission ($t = 0$). Further samples were collected 2 and 4 hours after the initial dose of penicillin if the neonate had not been delivered. Maternal GBS status was monitored as a function of time from onset of treatment in all cases before the second dose of penicillin.

The GBS cultures were performed on selective Todd-Hewitt medium supplemented with gentamicin and nalidixic acid (Lim Broth, Becton Dickinson, Sparks, MD, USA), in accordance with CDC recommendations [4]. The samples were incubated for 18–24 hours at $35 \pm 2^\circ\text{C}$ and subculture was performed on blood agar and Granada medium under environmental and anaerobic atmosphere conditions, respectively. After overnight incubation period, the plates were analyzed for GBS, and those on which microorganisms were not identified were re-incubated and analyzed at 48 hours.

Penicillin concentration was measured in the umbilical cord and amniotic fluid in relation to time from the beginning of treatment. Cord blood samples were collected from all participants after delivery. Amniotic fluid samples were obtained only from patients who delivered by cesarean without membrane rupture, and a sterile syringe was used to avoid sample contamination immediate to hysterotomy. The concentration of penicillin G was measured by high-performance liquid chromatography and ultraviolet detection. An internal standard (carbamazepine epoxide) was added to each sample, which was then extracted with organic solvent and passed through a reversed-phase column. The diluted solvent fraction was evaporated to dryness, resuspended in mobile phase and injected onto the high-performance liquid chromatography instrument. The concentration of penicillin in the samples was calculated by comparing the peak areas obtained (standardized via the internal standard) with a calibration curve and the appropriate controls.

The study was based on convenience sampling, whereby consecutive pregnant women during the study period who agreed to participate by providing informed consent were recruited. A maximum of 60 patients was recruited on the basis of a previous study on a similar topic [9]. Data were analyzed by Stata version 9.0 (Stata Corporation, College Station, TX, USA). Continuous and categorical data were expressed as mean $\pm$ SD and proportions, respectively. For GBS vaginal status, cumulative proportions were reported in association with the time interval since IAP administration. Penicillin G concentrations in amniotic fluid

and cord blood samples were also considered relative to the time interval (in minutes) between the initial IAP administration and delivery.

3. Results

The mean age of the study patients was 24.5 years (range 16–40). The mean gestational age at delivery was 39.4 weeks (range 37–41). The clinical and demographic characteristics of the study patients are listed in Table 1. Among the 60 study participants, only 35 (58%) received at least 4 hours of IAP. The time between administration of the first dose and delivery ranged from 2.0 to 11.5 hours (median 4.1 hours).

Among the 60 patients who were positive for GBS during prenatal screening and recruited for the study, 43 (72%) had a positive rectovaginal sample before initiation of IAP. Among these women, the first 2 hours of IAP were sufficient for the culture to become negative for 23 (53%). By 4 hours, only 5 (12%) remained positive for GBS (Fig. 1).

The penicillin G concentration in cord blood samples ranged from 0.5 to $35.0 \mu\text{g/mL}$ (mean $2.4 \pm 10.6 \mu\text{g/mL}$). In all 60 cases, the serum level of penicillin G in the newborn was above the MIC for GBS ($0.12 \mu\text{g/mL}$).

Among the eight women who delivered by cesarean, the penicillin G concentration ranged from 0.5 to $4.6 \mu\text{g/mL}$ (mean $2.55 \pm 1.66 \mu\text{g/mL}$). The concentration of penicillin G in amniotic fluid was significantly above the MIC in samples from the seven women for whom surgery was completed within 4 hours of IAP initiation, but before 195 min, it rose more slowly than in the cord blood compartment (Table 2, Fig. 2).

4. Discussion

In the present study, 42% of patients did not complete 4 hours of IAP before delivery. This finding is similar to that in a previous report [8]. However, a subset of the present data showed that less than 4 hours of IAP infusion effectively achieved levels of penicillin G above the MIC for GBS in both the neonatal bloodstream and amniotic fluid, suggesting that, from a pharmacological point of view, this treatment might yield benefits even in a precipitous labor.

However, from a pharmacokinetic point of view, β -lactam antibiotics exhibit maximal clinical benefit (bactericidal activity) in a time-dependent manner, and antibiotic concentrations in fluids and tissues do not necessarily represent the maximum effect in a given moment [13]. In the present study, 4 hours of IAP was required to reduce the

Table 1
Clinical and demographic characteristics of the study patients.^a

Characteristics	Patients (n = 60)
Age, y	24.5 $\pm$ 6.2
BMI	29.2 $\pm$ 5.2
Gestational age at delivery, wk	39.4 $\pm$ 1.1
Parity	
0	28 (47)
≥ 1	32 (53)
Ethnic origin	
White	50 (83)
Native (Aboriginal)	7 (12)
Black	3 (5)
Mode of delivery	
Vaginal	48 (80)
Forceps	4 (7)
Cesarean	8 (13)
No. of samples	
GBS rectovaginal culture ^b	60
Cord blood ^c	60
Amniotic fluid ^d	8

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by the square of height in meters); GBS, group B streptococcus.

^a Values are given as mean $\pm$ SD or number (percentage), unless indicated otherwise.

^b From GBS-positive women in labor before starting IAP and then every 2 hours.

^c Collected at time of delivery.

^d Only from cesarean delivery.

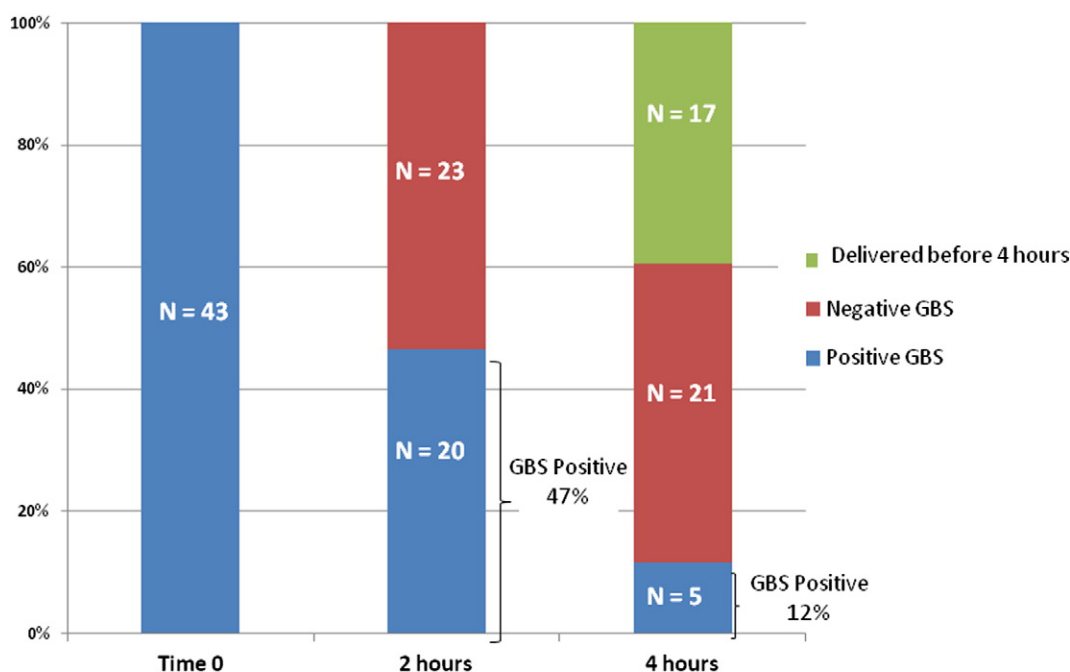


Fig. 1. Change in GBS carrier status of participants during intrapartum antibiotic prophylaxis treatment (n = 43). Abbreviation: GBS, group B streptococcus.

Table 2
Penicillin in amniotic fluid and cord blood among women delivering by cesarean (n = 8).

Time between start of intrapartum antibiotic prophylaxis and delivery, min	Penicillin concentration	
	Amniotic fluid, µg/mL	Cord blood, µg/mL
120	0.5	2.5
140	0.9	1.7
165	2.1	4.3
195	5.2	5.6
210	2.2	2.6
215	3.6	3.3
240	2.9	1.1
370 ^a	4.6	1.0

^a After second dose of penicillin G.

proportion of women with positive GBS cultures to 12%. This finding is consistent with the results of Turrentine et al. [14] and Fairlie et al. [15], who suggested that the incidence of neonatal clinical sepsis is increased in newborns delivered by GBS-positive mothers who are exposed to less than 4 hours of IAP.

The main strengths of the present study are that changes in GBS carrier status were evaluated during the first 4 hours of IAP and that the penicillin G levels were measured in the fetus and amniotic fluid compartment for a subgroup of the study population. As far as we know, most previous studies have examined penicillin G serum concentrations during pregnancy (i.e. fetal serum or amniotic fluid) but not GBS colonization in the maternal genital tract during IAP. Other important strengths are that all laboratory procedures were

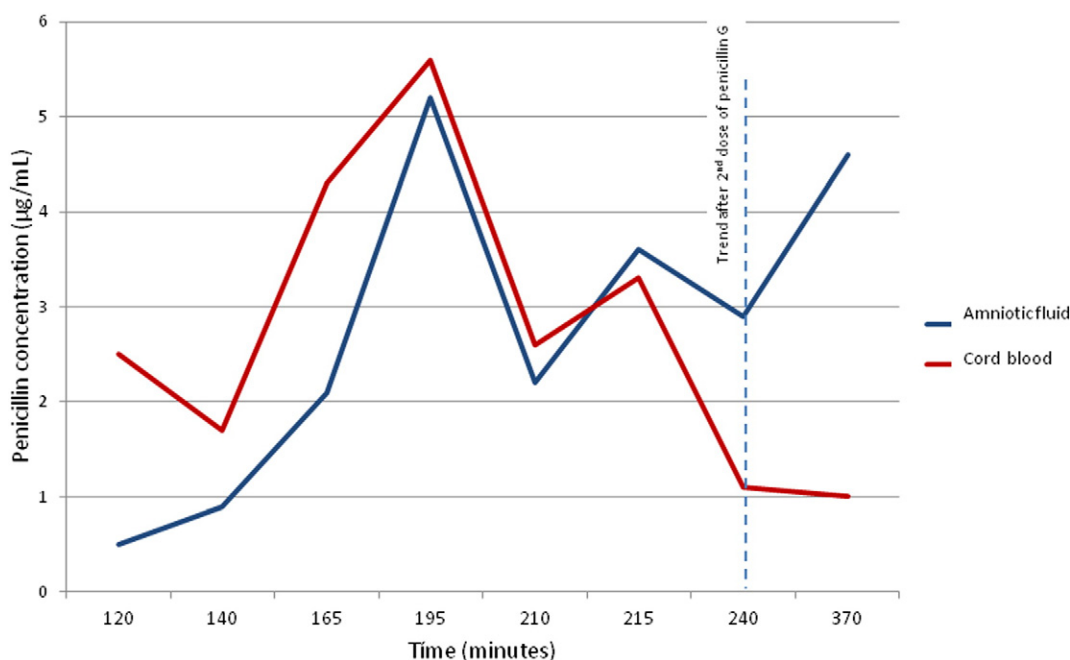


Fig. 2. Concentration of penicillin in amniotic fluid and cord blood among pregnant women who delivered by cesarean (n = 8).

done in line with CDC recommendations and quality control of data was strictly performed.

Nevertheless, the study has some limitations. The main results were based on intermediate outcomes such as GBS status, and the sample size was not large enough to evaluate neonatal outcomes. To determine the optimal duration of IAP, it would be necessary to correlate treatment duration with clinical outcome; however, early-onset GBS sepsis occurs rarely, especially in the setting of IAP. A well-designed, adequately sized and double-blind controlled trial would be needed to fully evaluate this clinical query [10].

A further limitation is that the present study described vaginal GBS status at 2 and 4 hours of IAP and the cumulative effect of antibiotics at these timepoints; however, it would be more informative to collect more samples at shorter intervals (e.g. every hour). In addition, precipitous deliveries were not included in the study, which was limited to pregnant women in the early active phase of labor for whom it was possible to collect at least one sample after IAP initiation. Therefore, the study is not exempt from selection bias.

In summary, the present data suggest that fetal serum and amniotic fluids levels of penicillin G far exceed the MIC before the 4 hours of IAP recommended by the CDC, suggesting that antibiotic prophylaxis might be effective in short periods and should be administered even in precipitous deliveries. However, the present findings also showed that at least 2 hours of treatment was necessary to achieve the goal of halving the incidence of GBS genital tract colonization. The maximum effect of IAP was observed at 4 hours just before the next IAP dose. As a result, newborns exposed to fewer than 4 hours of IAP should be considered inadequately protected for the purposes of neonatal management. In conclusion, and in agreement with Turrentine [16], clinicians and institutions might attempt simple strategies to begin IAP as soon as possible to reach the recommended 4 hours of IAP and to obtain maximal clinical benefit.

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Conflict of interest

The authors have no conflicts of interest.

References

- [1] American College of Obstetricians and Gynecologists Committee on Obstetric Practice. ACOG Committee Opinion No. 485: Prevention of early-onset group B streptococcal disease in newborns. *Obstet Gynecol* 2011;117(4):1019–27.
- [2] Anthony BF, Okada DM, Hobel CJ. Epidemiology of group B Streptococcus: longitudinal observations during pregnancy. *J Infect Dis* 1978;137(5):524–30.
- [3] Verani JR, Schrag SJ. Group B streptococcal disease in infants: progress in prevention and continued challenges. *Clin Perinatol* 2010;37(2):375–92.
- [4] Verani JR, McGee L, Schrag SJ. Division of Bacterial Diseases, National Center for Immunization and Respiratory Diseases, Centers for Disease Control and Prevention (CDC). Prevention of perinatal group B streptococcal disease—revised guidelines from CDC, 2010. *MMWR Recomm Rep* 2010;59(RR-10):1–36.
- [5] Boyer KM, Gotoff SP. Prevention of early-onset neonatal group B streptococcal disease with selective intrapartum chemoprophylaxis. *N Engl J Med* 1986;314(26):1665–9.
- [6] Schrag SJ, Zywicki S, Farley MM, Reingold AL, Harrison LH, Lefkowitz LB, et al. Group B streptococcal disease in the era of intrapartum antibiotic prophylaxis. *N Engl J Med* 2000;342(1):15–20.
- [7] Lin FY, Brenner RA, Johnson YR, Azimi PH, Philips III JB, Regan JA, et al. The effectiveness of risk-based intrapartum chemoprophylaxis for the prevention of early-onset neonatal group B streptococcal disease. *Am J Obstet Gynecol* 2001;184(6):1204–10.
- [8] Barber EL, Zhao G, Buhimschi IA, Illuzzi JL. Duration of intrapartum prophylaxis and concentration of penicillin G in fetal serum at delivery. *Obstet Gynecol* 2008;112(2 Pt 1):265–70.
- [9] McNanley AR, Glantz JC, Hardy DJ, Vicino D. The effect of intrapartum penicillin on vaginal group B streptococcus colony counts. *Am J Obstet Gynecol* 2007;197(6):583.e1–4.
- [10] Ohlsson A, Shah VS. Intrapartum antibiotics for known maternal Group B streptococcal colonization. *Cochrane Database Syst Rev* 2014;6:CD007467.
- [11] de Cueto M, Sanchez MJ, Sampedro A, Miranda JA, Herruzo AJ, Rosa-Fraile M. Timing of intrapartum ampicillin and prevention of vertical transmission of group B streptococcus. *Obstet Gynecol* 1998;91(1):112–4.
- [12] Davis RL, Hasselquist MB, Cardenas V, Zerr DM, Kramer J, Zavitskovsky A, et al. Introduction of the new Centers for Disease Control and Prevention group B streptococcal prevention guideline at a large West Coast health maintenance organization. *Am J Obstet Gynecol* 2001;184(4):603–10.
- [13] Sinnollareddy MG, Roberts MS, Lipman J, Roberts JA. β -Lactam pharmacokinetics and pharmacodynamics in critically ill patients and strategies for dose optimization: a structured review. *Clin Exp Pharmacol Physiol* 2012;39(6):489–96.
- [14] Turrentine MA, Greisinger AJ, Brown KS, Wehman OA, Mouzoon ME. Duration of intrapartum antibiotics for group B streptococcus on the diagnosis of clinical neonatal sepsis. *Infect Dis Obstet Gynecol* 2013;2013:525878.
- [15] Fairlie T, Zell ER, Schrag S. Effectiveness of intrapartum antibiotic prophylaxis for prevention of early-onset group B streptococcal disease. *Obstet Gynecol* 2013;121(3):570–7.
- [16] Turrentine M. Intrapartum antibiotic prophylaxis for Group B Streptococcus: has the time come to wait more than 4 hours? *Am J Obstet Gynecol* 2014;211(1):15–7.